DAY 09 | PRE-READ

Big Lists and a New Language

A light read before Day 9. About 15 minutes, no code required.

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Why read this?

Real apps have thousands or millions of records. Tomorrow you will learn how to summarise them, show them in pages, and find things fast. In the afternoon you will meet JavaScript, the language that runs inside every web browser.

1. Asking the database for totals

Imagine a shop manager who wants total sales per category. They could photocopy every receipt and add them up at home, or ask the accountant, who already has all the receipts, to send just the totals. Asking the database to calculate is the second approach. MongoDB calls this **aggregation**.

2. Pages, not a giant scroll

Search results show 10 items per page, not a million. This is **pagination**. It keeps apps fast and screens readable. Some apps use page numbers; social feeds use "load more" as you scroll.

3. The index at the back of a book

To find "photosynthesis" in a biology book, you do not read every page. You check the index and jump to page 142. A database **index** works the same way. It makes searching fast, at the cost of a little extra space and effort when adding new data.

Key idea: totals in the database, pages for the screen, indexes for speed.

4. Meet JavaScript

Python runs on servers and laptops. JavaScript runs in every web browser, and with a tool called Node.js it also runs on servers. If you know Python, JavaScript will feel familiar: it has variables, functions, lists (called arrays) and dictionaries (called objects). The main visible difference is that JavaScript uses curly braces { } instead of indentation.

Idea	Python	JavaScript
A list of things	list	array
Labelled information	dict	object
Print something	print()	console.log()
Group a block of code	indentation	{ curly braces }

5. Think about these before class

- What summary numbers would a manager want from an online shop every morning?
- Which apps you use show page numbers, and which use endless scrolling?
- Open any website, right-click and choose Inspect, then the Console tab. That is where JavaScript runs.

Hajimemashite (はじめまして): nice to meet you, said when meeting someone new, like JavaScript